

Sher Partap Singh

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Education

Plaksha University

B.Tech. Computer Science and Artificial Intelligence; Minor: Technology Entrepreneurship

Mohali, Punjab, India

Expected May 2027

—CGPA: 8.97/10 (85.2% equivalent); Relevant coursework: Advanced NLP, Reasoning with LLMs, Machine Learning and Pattern Recognition, Deep Learning, Reinforcement Learning, Intelligent Machines, Data Structures and Algorithms.

Pratap Public School

CBSE Class XII: 88%; SAT Math: 780

Karnal, Haryana, India

May 2023

Research Interests

Healthcare AI and longitudinal clinical intelligence; trustworthy LLM applications and hallucination detection; multi-agent systems; reinforcement-learning robustness; AI for education.

Research Experience

Research Intern — Prof. Sunita Chauhan

Plaksha University

Dec 2025 – May 2026

Mohali, India

- Built and evaluated LLM-assisted pipelines for longitudinal radiology understanding, paired-report extraction, RECIST-oriented oncology progression tracking, and queryable fact-graph representations.
- Explored OCR pipelines for handwritten clinical notes and contributed to temporal reasoning workflows tracking progression, stability, worsening, and resolution across reports.

Business Analyst Intern (AI Committee) — Flipkart

Analytics and AI initiatives in large-scale e-commerce systems

May 2026 – Jul 2026

Bangalore, India

- Worked on business analytics, applied AI deployment, and internal AI strategy discussions within large-scale e-commerce data systems.

Student Intern — Prof. Anish Chowdhury

Plaksha University

Jun 2025 – Aug 2025

Mohali, India

- Built a React/Tesseract.js Curriculum Gap Analysis Tool using Azure OpenAI embeddings; identified syllabus gaps for lateral-entry students 10× faster than manual comparison.
- Developed a Streamlit analytics agent generating visualizations, statistical summaries, and natural-language insights from CSV/Excel files.

Research Assistant — Lab for Economic Behaviour in Organisations

Plaksha University

Jun 2024 – Sep 2024

Mohali, India

- Conducted large-scale research-data scraping with Python, Selenium, BeautifulSoup, and proxy rotation; preprocessed datasets and synthesized literature reviews.

Publications and Selected Research

The Fact Graph: A Unified Ontological Representation for Multi-Modal Clinical Intelligence

- Accepted at IntelliSys 2026 (Springer SVPROC). Contributed to longitudinal radiology intelligence, paired-report extraction, temporal reasoning, and fact-graph architectures for clinical event timelines.

SAC's Entropy Coefficient as an Implicit Success Signal in Robotic Manipulation

- Accepted poster at CAISc 2026; sole author. Analyzed 128 training runs across five Meta-World tasks; identified entropy-coefficient regimes that classify 23/24 seeds (96%) on the hardest tasks and reduce final returns by up to 98% under fixed entropy annealing.

Selected Projects

Bugzer — AI Multi-Agent Web QA System | Python/FastAPI, Next.js, Docker, GCP

Jan 2025 – Present

- Built browser agents that discover bugs and generate reports from console logs, network traces, latency metrics, and screenshots, with LLM-assisted debugging recommendations.

Quad — Educational Platform | Next.js, Supabase, PostgreSQL, TypeScript

Dec 2024 – Mar 2025

- Developed scheduling, calendar synchronization, authentication, and video-management workflows for a Bharti Airtel Foundation Hackathon platform used by 100+ schools.

Bayesian Occupancy Grid Mapping | Python, Bayes' Filter, Matplotlib

2024

- Built and visualized real-time occupancy maps from noisy IR-sensor measurements after each robotic action.

Technical Skills and Leadership

Languages: Python, C++, C, MATLAB, JavaScript/TypeScript, SQL, Shell. **AI/ML:** Machine Learning, NLP, LLM Applications, Clinical NLP, Bayesian Inference, OCR, Reinforcement Learning.

Tools: PyTorch, HuggingFace Transformers, FAISS, FastAPI, React, Next.js, Node.js, Streamlit, Docker, GCP, Azure OpenAI, OpenAI API, Selenium, BeautifulSoup, Git, Linux.

Leadership: Plaksha AI Committee member (Jan 2026 – Present); Junior Teaching Assistant, Young Technology Scholars Program (May – Jun 2025, mentored 80+ students); Founder, Wings of Fire Club (2022 – Present).

Achievement: 1st Place, Predli AI Case Study Competition (2025); Full Scholarship Recipient, Plaksha University.